

Tender Submission

Feeder Balancing Module (FBM) Tool — Design, Build, Host, Support and Maintain

Enquiry / Tender Number	3239CXMWP
Tender Title	Feeder Balancing Module (FBM) Tool
Issued By	Eskom Holdings SOC Ltd — Distribution Group IT
Bidder	T2 Technologies
Contract Form	NEC3 Professional Services Contract (Option A), ~6.5-year term
Closing Date & Time	10 September 2026 at 10:00 (South African Standard Time)
Bid Validity	[PLACEHOLDER: validity period per ITT, e.g. 120 days from closing]
Classification	Confidential — Tender Response

Document Control

Item	Detail
Document Title	T2 Technologies — FBM Tool Tender Submission
Version	1.0 (Draft for tender-specific completion)
Prepared For	Eskom Holdings SOC Ltd — Distribution Group IT
Prepared By	T2 Technologies
Enquiry Number	3239CXMWP
Classification	Confidential
Date	[PLACEHOLDER: submission date]

Completion note: Replace every [PLACEHOLDER: ...] with the exact wording and values from the issued tender pack before submission. Where this document and the issued tender pack conflict, the tender pack takes precedence. All company registration documents, tax compliance status, B-BBEE evidence, key-personnel CVs, priced schedules and signed returnable forms must be attached in the format prescribed by Eskom.

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Letter of Tender Submission

To: The Bid Evaluation Committee

Organisation: Eskom Holdings SOC Ltd — Distribution Group IT

Enquiry: 3239CXMWP — Feeder Balancing Module (FBM) Tool

Dear Sir / Madam,

T2 Technologies is pleased to submit this proposal for the design, build, hosting, support and maintenance of the Feeder Balancing Module (FBM) Tool. We understand that Eskom Distribution is not asking for an isolated software build. Eskom needs a governed, auditable capability that consolidates fragmented source-system data into a single trusted universe, lets energy analysts correct network configuration and allocate unallocated sales, calculates reliable energy-balancing results, and publishes those results and exception reports back into the business so that losses can be found, explained and reduced.

Our response is deliberately **evidence-led**

. Rather than describe an intended tool in the abstract, we have already built a

working FBM prototype

that demonstrates the interactive core of the scope — the OU hierarchy, network mapping, technical-loss and kWh adjustments, the balancing calculation, and a loss heat map with adjustable conditional formatting. This de-risks the most heavily weighted parts of the evaluation because the committee can see the functionality operate, not merely read about it.

T2 Technologies is a South African technology company delivering dependable end-to-end digital transformation. The company is 100% black-owned and operates with Level 1 B-BBEE credentials, subject to inclusion of the current valid supporting certificate or affidavit in the final bid pack. Our delivery model emphasises local capability development, accountable governance under the NEC3 contract, and practical transfer of technical knowledge to Eskom personnel.

We confirm our commitment to the requirements of this enquiry and to working collaboratively with Eskom to finalise the implementation plan, milestones, service levels and transition arrangements in accordance with the contract.

Yours faithfully,

[PLACEHOLDER: Authorised Signatory] | [Position]

T2 Technologies | [Email] | [Mobile]

1. Executive Summary

Feeder balancing reconciles **energy delivered** (measured by statistical / MV90 meters) against **energy consumed** (customer meters billed in CC&B), after accounting for **technical losses**

incurred in physically transporting electricity. When delivered energy is compared with consumption plus technical loss, any residual is

non-technical loss (NTL)

— the number Eskom Distribution needs to find, explain and reduce. Today this process is constrained by fragmented source data, manual screening, manual network configuration and manual sales allocation, which limits both accuracy and the speed of corrective action.

T2 Technologies proposes a secure, modular FBM platform that automates data consolidation into a single BI universe, provides a user-friendly application to correct network configuration and allocate unallocated CDU sales, calculates energy-balancing results with adjustable technical-loss percentages, and publishes results, heat maps, charts and the full suite of exception reports. The platform integrates to Eskom's work- and asset-management systems to raise work requests on faulty metering and to initiate audits on the worst-performing networks.

Why T2 Technologies

- **Proven, not promised.** A working FBM prototype already demonstrates the interactive configuration, adjustment, calculation and heat-map surfaces that carry the largest functional weighting.
- **Eskom-aligned architecture.** Designed for the Eskom Azure tenant with data sovereignty in South Africa, SSO/MFA against Eskom identity, RBAC, AES-256 at rest and TLS 1.3 in transit, and integration to Eskom's ESB/iPaaS estate.
- **Security and compliance by design.** SOC 2 Type II posture where hosted as a cloud service, SAST/DAST and penetration testing before production, POPIA compliance, WAF and DDoS protection, and 24-hour breach notification.
- **End-to-end delivery.** From discovery and the BI data platform through configuration, calculation, reporting, publishing, integration, testing, go-live and long-term support under NEC3.
- **Local skills transfer.** Documentation, paired delivery and administrator training reduce Eskom's long-term dependency on the supplier and support transformation objectives.

Evaluation-aware summary: This response is structured around Eskom's five weighted evaluation areas — Key Requirements (20%), Functional (40%), Non-Functional (20%), Cloud (5%) and Security (15%) — and is written to meet or exceed the functionality threshold. Section 3 maps each evaluation area to the section that answers it.

2. Understanding of Eskom's Requirement

Eskom requires a like-for-like replacement of the existing FBM tools with defined enhancements and integration, delivered as a governed digital capability. The tool must consolidate source-system data, enable correction of network configuration and sales allocation, compute reliable balancing results, and publish results and reports for loss management and targeted audits.

2.1 From As-Is to To-Be

As-Is (today)	To-Be (required)
Source data flows partially automated to BI; MV90.EU file still manual	Fully automate all source feeds (incl. MV90.EU) into one production BI universe
Manual screening of data for challenges	Advanced source-data analytics and data mining with anomaly detection
Manual network configuration; reliance on field resources	Application-driven mapping of stats meters to feeder/substation configurations, with CNL-assisted linking
Manual allocation of unallocated CDU/PP sales	Tools to allocate unallocated sales by percentage to dense prepaid networks
Manual kWh manipulation; single default technical-loss %	Auditable kWh adjustments and per-feeder technical-loss % that becomes the new default
OU-based results, manually combined for national view	National consolidation with drill-down Cluster/OU/Zone/Sector/CNC/Meter
Reports produced in the legacy tool	The 28 report extracts, heat maps and charts developed in the BI environment and published
No versioned storage of monthly mapping	Every data iteration and mapping stored, time-stamped and reusable next month

2.2 The reconciliation at the heart of the scope

For every feeder, the platform summates customer sales on the mapped feeder and compares them against the consumption measured by the mapped stats meter, incorporating the feeder's technical-loss percentage. The result is the first-version loss, refined through validated kWh adjustments and sales allocation. Results roll up through the OU hierarchy to national level and down to feeder and stats-meter level, and are published to the corporate BI warehouse and shared folder with meter statuses set accordingly.

3. Response Map to the Evaluation Criteria

The table below maps each weighted evaluation area to the section of this response that addresses it, so the committee can score efficiently. Weightings reflect the Technical Evaluation Criteria (TEC); please verify against the issued TEC before finalisation.

Evaluation area	Weight	Answered in
Key Requirements (referenceability & standing)	20%	Section 4 + Annexures (references, B-BBEE, registration)
Functional Requirements (BRS 1–17, RQ, reports)	40%	Section 5 + Section 12 (working prototype) + Section 20 (matrix)
Non-Functional Requirements	20%	Section 6 + Section 13/14 (testing, support, BCP)
Cloud	5%	Section 7 + Section 9 (architecture)
Security	15%	Section 8

Threshold awareness: Eskom applies a minimum functionality threshold (typically 70%) before price and B-BBEE are considered. Every functional and non-functional line item in Section 20 is answered with a Comply / Partial / Comply-by-delivery position and a cross-reference to the supporting evidence.

4. Key Requirements Response (20%)

The Key Requirements area assesses the bidder's standing and ability to demonstrate comparable, verifiable delivery. T2 Technologies responds as follows; all claims are supported by the annexures listed and must be finalised with current evidence before submission.

Key requirement	T2 Technologies response	Evidence
Referenceability (verifiable comparable projects)	T2 presents comparable delivery in data integration, transaction/energy data reconciliation and enterprise platform delivery, supported by signed reference letters and contactable referees.	Annexure I: [PLACEHOLDER: 3+ signed reference letters]
Company registration & legal standing	Registered South African company in good standing.	Annexure B: [PLACEHOLDER: CIPC registration]
Tax compliance	Valid tax compliance status.	Annexure C: [PLACEHOLDER: Tax PIN / status]
B-BBEE	100% black-owned; Level 1 contributor.	Annexure D: [PLACEHOLDER: valid B-BBEE certificate/affidavit]
Financial standing	Financial capacity to deliver a ~6.5-year NEC3 engagement.	Annexure F: [PLACEHOLDER: banking / financials if required]

Integrity note: Only references supported by signed appointment letters, purchase orders, completion certificates or contactable referees should be presented as formal contractual references. Company experience, individual team-member experience and internal product development must be clearly distinguished.

5. Functional Requirements Response (40%)

This section responds to every Business Requirement (BRS 1–17) and the enhancement requirements (RQ), and identifies which capabilities are already demonstrable in the working prototype (Section 12). Legend: **Comply** = available/demonstrable now;

Comply by delivery

= delivered within the implementation plan;

Partial (prototype)

= interactive behaviour demonstrable on sample data, completed on real data during delivery.

5.1 Data platform (BRS 1–4)

BRS	Requirement	Position	How we deliver
1	Import & consolidate all source systems (MV90 metering, CC&B sales, unallocated CC&B, Small World/Maximo network, CNL) into one BI universe	Comply by delivery	Automated ETL/iPaaS feeds into a governed BI FBM universe; completes the outstanding MV90.EU! production automation
2	Advanced source-data analytics and data mining with anomaly detection	Comply by delivery	Rule-based anomaly detection with user follow-up and custom analyst queries across combined datasets
3	Upload data from BI into a user-friendly configuration/allocation application	Comply by delivery	Governed load path from BI into the FBM application
4	Manage current & historical meter/feeder statuses; filter, modify per rules, rename meters	Comply by delivery	Status model with history, filtering, rule-based modification and audit

5.2 Configuration, allocation & calculation (BRS 5–10)

BRS	Requirement	Position	How we deliver
5	Configure the network by mapping stats meters to feeder/substation configurations (CNL-assisted)	Partial (prototype)	Interactive tree + org-chart mapping with drag-and-drop and clustering is demonstrable now; wired to real data + CNL during delivery
6	Allocate unallocated CDU/PP sales by percentage to dense prepaid networks	Partial (prototype)	CDU allocation panel demonstrable now; real unallocated-sales feed during delivery
7	kWh adjustments (energy delivered and/or consumption, up/down)	Comply (prototype)	Per-feeder delivered & sales adjustments implemented; audited and persisted on real data
8	Adjust technical-loss % (10% default; per-feeder override becomes the new default)	Partial (prototype)	Global default + per-feeder override implemented; persistence of the new default during delivery
9	Calculate balancing results (sum sales per feeder vs measured consumption)	Comply (prototype)	Balancing engine runs across the hierarchy now; validated on real inputs during delivery
10	Display losses per OU hierarchy with drill-down to feeder & stats-meter	Partial (prototype)	Results dashboard + drill-down demonstrable; full national roll-up during delivery

5.3 Reporting, publishing & storage (BRS 11–16)

BRS	Requirement	Position	How we deliver
11	28 specified report extracts (pivot grids) — see 5.5	Comply by delivery	All 28 extracts developed in the BI environment after data acceptance and publishing
12	Heat maps with adjustable conditional colour-coding per feeder & OU	Comply (prototype)	Loss heat map with adjustable NTL%/kWh thresholds is demonstrable now
13	Charts: bar (current month) + line (historical), kWh delivered vs used and NTL kWh/% per OU! substation	Partial (prototype)	Balancing charts demonstrable; historical series added with stored iterations
14	Publish balanced results to online DB/BI warehouse + corporate shared folder; set meter statuses	Comply by delivery	Publishing pipeline to corporate BI + shared folder with meter-status setting
15	Display KPI results consolidated nationally with drill-down Cluster/OU/Zone/Sector/CNC/Meter	Comply by delivery	National consolidation with RBAC-scoped drill-down
16	Store latest data mapping + every iteration, time-stamped, reusable next month	Comply by delivery	Versioned storage of every data iteration and mapping in BI

5.4 Device management & enhancements (BRS 17, RQ)

Ref	Requirement	Position
17	Meter-reading automation; log WO to maintenance system for faulty meters	Comply by delivery
RQ2	Stats-meter readings managed via online Oracle DB; Stats Meter Management Module to automate MV90	Comply by delivery
RQ4	Upgrade local DB to SQL Server/MySQL/PostgreSQL (cater for Eskom PPU customers)	Comply by delivery
RQ5	CNL data mapping & validation module with graphical non-conformance overview	Comply by delivery
RQ7	FBM data-validation module (continuous, rule-based, role-routed exceptions)	Comply by delivery
RQ8	FBM meter-status flagging from MV90 status (multi-level)	Comply by delivery
RQ10/12/13/16	Additional exception & analysis reports (no-delivery feeders, supply points, customer/feeder linking, all CC&B customers)	Comply by delivery
RQ17/18	Business levels based on feeders (not substations); substations shared across OU boundaries	Comply by delivery
RQ19/20	Dx substation balancing; MTS balancing (Tx/Dx incoming vs Dx outflow reconciliation)	Comply by delivery
RQ21	Three published reports: Bulk feeders, Reticulation feeders, Consolidated	Comply by delivery

5.5 The 28 report extracts (BRS 11)

All 28 extracts will be delivered in the BI environment: stats-meter analysis per meter; feeder mapping scenarios; Feeder Consumption Analysis (Fdr/Trfm/Cust); Feeder Consumption Analysis (Fdr/Cust); new stats meters imported; stats meters previously imported now missing; new network locations imported; network locations previously imported now missing; network locations previously mapped no longer mapped; stats meters previously mapped no longer mapped; current vs previous stats-meter import comparison; transformers in engineering not in CC&B; premises with kWh but no Account ID; conventional transformers in CC&B not in engineering; PPU transformers in CC&B not in engineering; feeders not mapped; feeders with no consumption; worst-performing feeders; transformers_bulk on feeder; trfms_bulk/consumption on feeders; indicators (business) current vs previous; indicators (meters/feeders) current vs previous; LPU status current vs previous; CNL LPUs in engineering not linked to CC&B; CNL LPUs in CC&B not in engineering; CNL Trfs & bulks with no sales; CNL duplicate premises; feeders & meters not mapped.

6. Non-Functional Requirements Response (20%)

Non-functional area	T2 Technologies commitment
Performance	Record save within ~1s and retrieval within ~3s under agreed load; throughput sized to national data volumes. Baselines validated by performance testing (Section 13).
Availability & scalability	High availability across Azure availability zones; horizontally scalable, containerised services sized for all OUs at national scale.
Backup & archiving	Incremental daily encrypted backups kept offsite; retention aligned to Eskom policy; every monthly data iteration archived and versioned.
Business continuity & DR	System criticality treated as Tier 0/1 (Safety & Revenue / Mission Critical); RTO within 8–24h and defined RPO; real-time replication to a DR site in a different SA region; DRP and restore procedures tested annually with results shared with Eskom Cyber Security.
Usability & accessibility	Role-based, task-focused UI consistent with the demonstrated prototype; accessible and responsive.
Maintainability & portability	Modular, open-standards architecture avoiding lock-in; documented interfaces and portable data.
Training & documentation	Administrator, analyst and service-desk training; architecture, API, security, deployment and operations documentation.
Project management	Delivery under a formal PM methodology with NEC3 governance, stage gates and RAID management (Section 10–11).

7. Cloud Response (5%)

- **Hosting model.** Preferred deployment within the **Eskom Azure tenant / VPC** (Platform-as-a-Service), keeping the database within the Eskom corporate LAN/BIN behind the perimeter firewall. Where any component is offered as an external cloud service, SOC 1/2 Type II attestation applies (Section 8).
- **Data sovereignty.** All data stored and processed within South Africa; DR region also within South Africa.
- **Composability.** Containerised microservices, infrastructure-as-code, and environment automation across DEV/DR/PROD.
- **Integration.** Supports Eskom's prevailing ESB / API / iPaaS estate (e.g. IBM App Connect, TIBCO, WSO2, webMethods, Azure Service Bus, Oracle Service Bus, MuleSoft, SSIS, Power Automate) for security, logging and monitoring across on-prem, hybrid and multi-cloud.
- **Portability.** Open standards preserve Eskom's ability to evolve the deployment model and avoid vendor lock-in.

8. Security Response (15%)

Control domain	Commitment (aligned to the Scope of Work security requirements)
Attestation	For any cloud service storing/processing financial or PII/IP data, valid SOC 1 and SOC 2 Type II reports submitted to Eskom; annual summaries and bridge letters as required.
Identity & access	Integration with Eskom's on-prem IdP and MFA for SSO; RBAC with least privilege; privileged-access controls and joiner-mover-leaver procedures.
Encryption	AES-256 at rest; TLS 1.3 (or later) in transit.
Logging & audit	Audit trails, user-administration and activity logs enabled, encrypted and access-restricted to administrators.
Data protection	PII masked in Sandbox/DEV; data minimisation and retention controls; controlled export.
Backups & DR	Incremental daily encrypted offsite backups; real-time replication to a DR site in a different SA region; annually tested DRP and restore procedures.
Vulnerability management	SAST + DAST, vulnerability assessment and penetration testing before PROD; all critical/high/medium findings remediated pre-PROD; summaries submitted to Eskom Cyber Security for acceptance; defined patch-management process.
Network & perimeter	Database behind the perimeter firewall (Eskom LAN/BIN on-prem or partner private network in cloud); WAF for internet-facing apps; DDoS protection for all databases; database security management tooling.
Privacy & incident response	POPIA (and GDPR where applicable) compliance; breach notification to Eskom within 24 hours; change notification within one month for significant business/platform/hosting changes.
e-Discovery	Capability to identify, collect and produce electronically stored information on request.

9. Solution Architecture and Design Principles

The target architecture will be confirmed during discovery against Eskom's standards. T2 applies the following layered design.

- **Experience layer** — analyst and administrator interfaces for configuration, allocation, calculation, dashboards and reporting (as demonstrated in the prototype).
- **Application/services layer** — network configuration, sales allocation, technical-loss management, balancing calculation, status management and validation services.
- **Integration layer** — authenticated APIs and ESB/iPaaS connectors to MV90, CC&B, Small World/Maximo, CNL, and work/asset management.
- **Data layer** — the BI FBM universe, versioned monthly iterations, the upgraded relational database (SQL Server/ PostgreSQL) and the publishing store.
- **Platform layer** — Azure tenant, containers, secrets management, backup and DR.
- **Operations layer** — CI/CD, monitoring, logging, alerting, vulnerability management and audit reporting.

Principle	Application to FBM
Modularity	Separate configuration, calculation, reporting and publishing so each can evolve independently.
API-first & integration	Controlled interfaces to source systems rather than direct coupling.
Security by design	Identity, RBAC, encryption, secrets and auditability from inception.
Traceability & auditability	Every configuration change, adjustment and allocation is versioned and attributable.
Resilience & data governance	HA/DR, and defined ownership, quality controls, retention and lineage for critical data.

10. Delivery Methodology and Implementation Plan

T2 proposes a phased, agile delivery model governed by NEC3 stage gates — balancing iterative product delivery with the traceability expected of an Eskom programme. Timings are indicative and reconciled to the issued milestones during mobilisation.

Phase	Indicative	Key activities	Exit outcome
0 — Mobilisation	Wks 1–2	Contract mobilisation, governance, access, security clearance, delivery plan, environments	Approved mobilisation pack & plan
1 — Discovery & Blueprint	Wks 2–6	Source-system assessment, requirements baseline, BI universe design, target architecture, security assessment	Baseline, architecture & backlog
2 — Data Platform	Wks 5–12	Database upgrade, BI feeds incl. MV90.EU automation, load path into the app	Trusted data flowing to the app
3 — Configuration & Calculation	Wks 10–20	BRS 5–10 on real data; CNL linking; status mgmt; versioned persistence	Balancing results on real data
4 — Reporting & Publishing	Wks 18–28	28 report extracts, heat maps, charts, publishing pipeline, national consolidation	Reports & KPIs published
5 — Advanced & Integration	Wks 26–34	Data mining, RQ17–21, device mgmt & WO integration, validation module	Enhancements live
6 — UAT, Readiness & Go-Live	Wks 32–40	End-to-end, performance, security testing, UAT, training, DR rehearsal, go-live & hypercare	Production service + BAU transition

11. Programme Governance and Proposed Team

Forum	Purpose	Cadence
Steering Committee	Executive decisions, scope/budget, strategic risk, milestone approval	Monthly / as prescribed
Programme Review	Progress, dependencies, RAID, upcoming milestones	Weekly
Architecture / Security Review	Design decisions, integration standards, security controls	Per major design
Sprint Ceremonies	Planning, stand-ups, demos, retrospectives	Per sprint
Service Review	SLA, incidents, problems, releases, improvement	Monthly post go-live

11.1 Proposed core team

- Executive Sponsor / Account Lead; Programme/Project Manager; Solution/Enterprise Architect.
- Business Analyst (energy/losses domain); Data Engineers (BI/ETL); Software Engineers; Integration Engineer.
- DevOps/Cloud Engineer; Cybersecurity Specialist; QA/Test Lead; UX Specialist; Service/Support Lead.
- CVs, qualifications, role allocation and availability attached as Annexure H and matched to the resource schedule.

12. Relevant Experience and the FBM Working Prototype

A key differentiator of this bid is that the interactive core of the FBM scope already exists as a working prototype, available for live demonstration to the committee. It validates our understanding of the domain and materially de-risks the functional evaluation.

Prototype capability	Scope requirement demonstrated
OU hierarchy & network mapping	BRS 5 — tree + org-chart mapping of meters!'transformers!'feeders!'substations, drag-and-drop and clustering
kWh & technical-loss adjustments	BRS 7 & 8 — per-feeder delivered/sales adjustments and technical-loss % override
CDU allocation	BRS 6 — allocation of unallocated sales to feeders
Balancing calculation	BRS 9 — delivered vs consumed + technical loss across the hierarchy
Loss dashboard & heat map	BRS 12 — Gauteng loss heat map with adjustable conditional formatting; BRS 10/13 drill-down and charts

The prototype currently runs on representative Gauteng sample data. Under the implementation plan it is wired to Eskom's real source systems, the BI universe and the upgraded database, then extended nationally.

12.1 Representative experience (to be evidenced)

- Electricity data integration & feeder analytics — mapping and reconciliation of network, feeder, transformer and meter datasets for energy-flow analysis.
- Enterprise integration & data engineering — secure APIs, ETL pipelines and reconciliation across multiple source systems.
- [PLACEHOLDER: named comparable projects with signed reference letters, scope, duration and contactable referees].

13. Testing and Quality Assurance

Test level	Purpose
Unit	Verify components and balancing business logic at code level
API / Integration	Verify source-system interfaces, data contracts and error handling
Data / Reconciliation	Validate transformation, completeness, duplicates and control totals against source
Functional	Verify BRS/RQ behaviour and user stories
Performance	Validate the ~1s save / ~3s retrieve targets and national-scale throughput
Security	SAST/DAST, vulnerability assessment and penetration testing before PROD
UAT	Eskom users confirm fitness for operational use
Operational readiness	Validate monitoring, backup, recovery, runbooks and go-live procedures

14. Service Management, Support and Maintenance

Following go-live, T2 transitions the FBM Tool into a supported service for the NEC3 term. Response and restoration targets are aligned to the tender SLA; the values below are indicative pending confirmation.

Severity	Definition	Indicative response
P1 — Critical	Production unavailable, balancing/publishing stopped, or severe security event	[PLACEHOLDER: e.g. 30 min]
P2 — High	Major degradation or key function unavailable	[PLACEHOLDER: e.g. 1 hr]
P3 — Medium	Limited degradation with workaround	[PLACEHOLDER: e.g. 4 bus. hrs]
P4 — Low / Request	Minor defect, service request or query	[PLACEHOLDER: e.g. 1 bus. day]

- Support model & escalation matrix; monitoring & alerting; backup/restore procedures.
- Release & change management; operations runbook & known-error records.
- Monthly service report covering SLA, incidents, problems, changes and improvement actions.

15. Knowledge Transfer and Change Enablement

- Joint design and implementation sessions with nominated Eskom technical staff.
- Architecture, API, security, deployment and operations documentation.
- Administrator, analyst and service-desk training; role-based operational guidance.
- Handover of source code, configuration and pipeline definitions per the contract's IP provisions.
- Defined knowledge-transfer acceptance criteria before transition to BAU.

16. Risk Management

Risk	Rating	Mitigation
Source-data quality / availability (esp. MV90.EU automation)	High	Early data discovery, profiling, reconciliation controls and rehearsal loads
Integration uncertainty across CC&B, Small World/Maximo, CNL	High	Interface inventory, ESB/iPaaS standards, proof-of-concept integrations
Scope volume at national scale	Med-High	Phased delivery, prioritised backlog, formal NEC3 change control
Security & compliance obligations	High	Secure SDLC, SAST/DAST + pen test, remediation gates before PROD
Eskom dependency delays (access, decisions)	Med-High	Dependency register, named owners, escalation dates, readiness checklists
Adoption & change	Medium	Stakeholder engagement, demonstrations, training and phased rollout

17. Socio-Economic Development (SD&L)

- Use of South African technical and professional resources wherever feasible.
- Structured participation of junior/developing resources under senior supervision.
- On-the-job knowledge transfer to Eskom personnel as measurable deliverables.
- Support for local suppliers/specialist partners where compliant with procurement rules.
- SD&L commitments completed on the prescribed SDL&I template and aligned to the exact scoring framework.
- [PLACEHOLDER: specific SD&L percentages/commitments per Annexure].

18. Assumptions, Dependencies and Exclusions

18.1 Assumptions

- Eskom provides timely access to stakeholders, source systems, documentation and environments.
- Existing system owners cooperate with agreed integration and testing activities.
- Eskom nominates authorised owners for requirements, architecture, security, data and UAT decisions.
- Security clearance and site-access processes are communicated before mobilisation.

18.2 Exclusions unless specifically priced

- Third-party licence and cloud subscription fees; hardware and data-centre facilities.
- Large-scale historical data cleansing beyond the agreed migration scope.
- 24x7 managed operations unless included in the pricing schedule.
- Independent certification or external penetration-testing fees unless expressly priced.

19. Commercial Proposal Framework

Pricing is completed strictly in the format prescribed by the FBM Pricing Schedule (Annexure L). The table below is an internal reconciliation view only; all values transfer to the official priced schedule.

Cost component	Pricing basis	Amount (excl. VAT)
Mobilisation & Discovery	Milestone	R [PLACEHOLDER]
Data platform (DB upgrade + BI feeds)	Milestone	R [PLACEHOLDER]
Configuration, calculation & reporting build	Sprint / deliverable	R [PLACEHOLDER]
Integration & advanced enhancements	Milestone	R [PLACEHOLDER]
Testing & go-live	Fixed	R [PLACEHOLDER]
Hosting (Azure/PaaS)	Monthly / annual	R [PLACEHOLDER]
Support & maintenance (NEC3 term)	Monthly / annual	R [PLACEHOLDER]
Total tendered price	As prescribed	R [PLACEHOLDER]

Validity, escalation, disbursements, VAT, payment milestones, retention and any guarantees are completed exactly as required by the tender conditions and NEC3 contract data.

20. Functional & Non-Functional Compliance Matrix

This consolidated matrix is the committee's scoring aid. Positions: C = Comply now (prototype/available); D = Comply by delivery; P = Partial (prototype on sample data, completed on real data).

Ref	Requirement	Pos.	Reference
BRS 1	Import & consolidate source systems into BI universe	D	\$5.1
BRS 2	Advanced analytics & data mining	D	\$5.1
BRS 3	Upload BI data into the application	D	\$5.1
BRS 4	Manage meter/feeder statuses (current & historical)	D	\$5.1
BRS 5	Configure network (meter/feeder/substation mapping)	P	\$5.2, \$12
BRS 6	Allocate unallocated CDU/PP sales	P	\$5.2, \$12
BRS 7	kWh adjustments	C	\$5.2, \$12
BRS 8	Adjust technical-loss % (new default)	P	\$5.2, \$12
BRS 9	Calculate balancing results	C	\$5.2, \$12
BRS 10	Display losses per OU hierarchy w/ drill-down	P	\$5.2, \$12
BRS 11	28 report extracts	D	\$5.3, \$5.5
BRS 12	Heat maps (adjustable conditional formatting)	C	\$5.3, \$12
BRS 13	Charts (bar + historical line)	P	\$5.3
BRS 14	Publish results + set meter statuses	D	\$5.3
BRS 15	National KPI consolidation + drill-down	D	\$5.3
BRS 16	Store versioned data & mapping	D	\$5.3
BRS 17	Meter-reading automation + WO logging	D	\$5.4
RQ2–RQ21	Enhancements (Oracle stats DB, DB upgrade, CNL, validation, feeder business levels, Dx/MTS balancing, split reports)	D	\$5.4
NFR	Performance, availability, scalability, backup, DR/BCP, training, PM	D	\$6
Cloud	Azure/PaaS, SA sovereignty, ESB/iPaaS integration	D	\$7
Security	SOC 2, SSO/MFA, RBAC, AES-256/TLS 1.3, SAST/DAST, POPIA, WAF/DDoS	D	\$8

21. Declarations and Acceptance

T2 Technologies declares that the information in the final submitted bid will be accurate to the best of its knowledge and that any qualifications, assumptions or deviations are clearly disclosed. Tender-specific declarations are signed on the prescribed forms by an authorised representative.

Declaration	Completion
Authorised signatory	[PLACEHOLDER: name & position]
Signature / Date / Place	_____ / _____ / _____
Bid validity confirmed	[PLACEHOLDER: yes / as required]
Conflict of interest	[Attach prescribed form]
Tax compliance / PIN	[Attach / insert]
Central Supplier Database	[Attach / insert]
B-BBEE evidence	[Attach current valid certificate/affidavit]

22. Returnable Schedules and Annexure Checklist

Annex	Document	Included
A	Official tender forms & returnable schedules (Form A, Schedule Q, Annexure K)	☐
B	Company registration documents (CIPC)	☐
C	Tax compliance documentation	☐
D	B-BBEE certificate / affidavit	☐
E	CSD / supplier registration evidence	☐
F	Banking / financial confirmation (if required)	☐
G	Company profile	☐
H	Key personnel CVs and qualifications	☐
I	Project references / completion certificates	☐
J	Technical architecture / solution diagrams	☐
K	Implementation project plan	☐
L	Risk register	☐
M	Priced schedule (Annexure L pricing)	☐
N	SLA / support model	☐
O	Security & POPIA / data-protection response	☐
P	Method statement (240-126469599) & SHE requirements	☐
Q	SDL&I template	☐

Final bid readiness check

- Every mandatory returnable is present and signed where required.
- Enquiry number and title are consistent across the entire bid pack.
- Pricing totals reconcile across all schedules.
- All named personnel match the CVs and resource schedule.
- All reference claims are supported by contactable referees.
- No [PLACEHOLDER] text remains in the final submission.
- Electronic filenames, page limits, portal upload rules and closing time verified.